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SYSTEMATIC REVIEW

Selective internal radiation therapy for hepatic metastases of uveal melanoma: a systematic review

[1]**HARRY ALEXANDER**, BMSc (Hons), MBChB, [2]**DANIEL WEN**, [2]**MICHAEL CHU**, MBChB, FRACS, PhD, [3]**CATHERINE HAN**, MBChB, FRACP, PhD, [4]**PETER HADDEN**, MBChB, FRANZCO, [5]**ROBERT THOMAS**, MB BS, MRCP, FRCR, EBIR and [2]**ADAM BARTLETT**, MBChB, FRACS, PhD

[1]Department of Anatomy and Medical Imaging, Faculty of Medical and Health Sciences, University of Auckland, Auckland, New Zealand
[2]Department of Surgery, Auckland District Health Board, Auckland, New Zealand
[3]Department of Pharmacology and Clinical Pharmacology, Faculty of Medical and Health Sciences, University of Auckland, Auckland, New Zealand
[4]Department of Ophthalmology, Faculty of Medical and Health Sciences, University of Auckland, Auckland, New Zealand
[5]Department of Radiology, Imperial College London, London, UK

Address correspondence to: Dr Harry Alexander
E-mail: *hale087@aucklanduni.ac.nz*

Objective: Uveal melanoma (UM) commonly metastasizes to the liver. Treatment usually consists of liver-directed therapies, such as selective internal radiation therapy (SIRT). This review aimed to assess the effectiveness and safety of SIRT for hepatic metastases from UM.
Methods: The study protocol is available at OSF (https://osf.io/vhyct/). EMBASE and MEDLINE were searched until July 2020, using terms related to SIRT and hepatic metastases from UM. Studies reporting outcomes of SIRT in patients with UM and at least one hepatic metastasis were included. Data on overall survival (OS), hepatic progression free survival (hPFS) or tumor response were collected. The Newcastle-Ottawa Scale (NOS) was used to assess risk of bias.
Results: 11 studies were included, reporting outcomes for 268 patients with hepatic metastases from UM. Most studies (n = 9, 81.8%) were retrospective. Disease control was achieved in 170 patients (67.5%) and the median OS from time of SIRT was 12.3 months. Median hPFS was 5.4 months. Low-grade side-effects were common but serious complications were infrequent. There were two treatment-related deaths. The median NOS score was 6 (moderate risk of bias).
Conclusion: SIRT appears to be a safe and effective treatment for patients with unresectable hepatic metastases from UM. The certainty of our results is unclear due to predominantly retrospective data with moderate risk of bias. Further prospective studies are required to explore the role of SIRT in UM.
Advances in knowledge: SIRT appears to be a safe treatment for patients with unresectable hepatic metastases from UM. Further prospective work is required.

INTRODUCTION

Uveal melanoma (UM) is the most common primary intra ocular malignancy in adults, with a mean age-adjusted incidence of 5.1 per million in the United States.[1] In Australia, the incidence of UM is significantly higher and has been previously estimated at more than eight cases per million.[2] This may be explained by demographic characteristics and lifestyle behavior.

At the time of diagnosis of UM, 99% of patients have no clinical or radiological evidence of extraocular disease.[3] However, up to 50% of patients with UM will ultimately develop distant metastases, despite successful treatment of the primary tumor.[4] The liver is the most common site of metastasis, affected in up to 90% of patients with metastatic UM.[5] Approximately 46% of patients with metastatic UM have liver-only disease.[6]

Patients with metastatic UM have poor prognosis, and overall survival (OS) is closely related to hepatic tumor control. Without treatment, the median survival for patients with UM and liver metastases is reported to be 4–5 months[4]. By comparison, patients with UM and only extra hepatic metastases have a median survival of 19–28 months[4].

No evidence-based treatment guidelines exist for metastatic UM. Currently, there are no standardized and effective systemic treatments available for metastatic UM. As OS is strongly linked to hepatic tumor control, liver-directed therapies have an important role in the treatment of metastatic disease. However, few patients are eligible for liver resection due to the number of metastases or their anatomical location.[7]

Other liver-directed therapies that can be considered for patients with hepatic metastases from UM, include trans-arterial chemo-embolization (TACE), hepatic artery infusion (HAI), isolated hepatic perfusion (IHP), radiofrequency ablation or immuno-embolization (IE).[8,9] Newer therapies such as checkpoint inhibition and CAR T-cell therapy are also under investigation and early studies appear promising.[10]

Selective internal radiation therapy (SIRT) is now an established alternative liver-directed treatment for patients with unresectable hepatic tumors, such as hepatocellular carcinoma or hepatic metastases from colorectal cancer.[11,12] As patients with primary intra ocular UM respond strongly to radiation therapy,[13] SIRT has been proposed as a potential treatment for patients with hepatic metastases from UM. SIRT is an appealing treatment because it is minimally invasive, typically well-tolerated, and does not require general anesthesia.

Several studies have examined the use of SIRT in patients with hepatic metastases arising from UM.[14–18] A previous systematic review examined the role of SIRT in patients with liver metastases from ocular or cutaneous melanoma.[19] However, metastases from the uveal tract respond differently to SIRT compared to metastases from the skin, and the effect of primary tumor location on treatment outcome was not examined by the study. As a result, the effect of SIRT on hepatic metastases arising solely from UM remains unknown. Multiple studies, including the first prospective trial of SIRT in hepatic metastases from UM, have also been published since 2017.[20]

The aim of this review was to assess the clinical efficacy and safety of SIRT for the treatment of hepatic metastases from UM. The primary outcomes of interest were tumor response, hepatic progression-free survival (hPFS) and OS following SIRT.

METHODS AND MATERIAL

Data sources and search string

The study protocol is available on Open Science Forum (https://osf.io/vhyct/). A systematic electronic search was conducted through the EMBASE and OVID MEDLINE databases from inception until 14 July 2020, according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.[21] A combination of keyword searches (.mp) and MeSH terms (/) related to SIRT and hepatic metastases UM were used.[8] The full search string is provided in Supplementary Material 1. Identified articles were limited to the English language.

Inclusion and exclusion criteria

Studies reporting the outcome of SIRT for patients with histologically confirmed UM and at least one hepatic metastasis were eligible for inclusion in this study. We included studies using SIRT as either initial or salvage therapy (*i.e.* with or without prior systemic or liver-directed therapy). Studies were included regardless of whether they included patients with extra hepatic metastases. Clinical trials and observational studies were eligible for inclusion in our review.

We excluded studies examining the use of SIRT in patients with hepatic metastases from cutaneous melanoma. Studies which reported the outcomes of SIRT in patients with a mixed denominator (*i.e.* both cutaneous and ocular melanoma) were also excluded, because metastases from UM behave differently to those from cutaneous melanoma. Case reports, non-published data and conference abstracts were excluded.

Data extraction

Title and abstracts were searched independently by two reviewers (HA and MC). Full texts were reviewed independently by two reviewers (HA and DW). Data of interest were collected onto a standardized spreadsheet. Discrepancies were resolved by discussion between the two reviewers (HA and DW), with involvement of a third author (MC) where necessary. Duplicates were excluded. Backward and forward citation chaining was conducted manually to identify any other potentially relevant studies.

Outcomes of interest

We collected data on study design and patient demographics (age, gender, performance status, extent of metastatic disease and hepatic tumor load). Data were also collected on treatment (pre-SIRT liver-directed and systemic treatment, SIRT procedure), and outcome (disease control, survival and toxicity). The primary outcomes were tumor response (according to RECIST criteria,[22] median hPFS and median OS (from time of SIRT and from diagnosis of hepatic metastases). Secondary outcomes were clinical and biochemical toxicity, graded according to Common Terminology Criteria for Adverse Events (CTCAE).[23] For studies with incomplete study details or outcome measures, the corresponding author was contacted via e-mail for additional data.

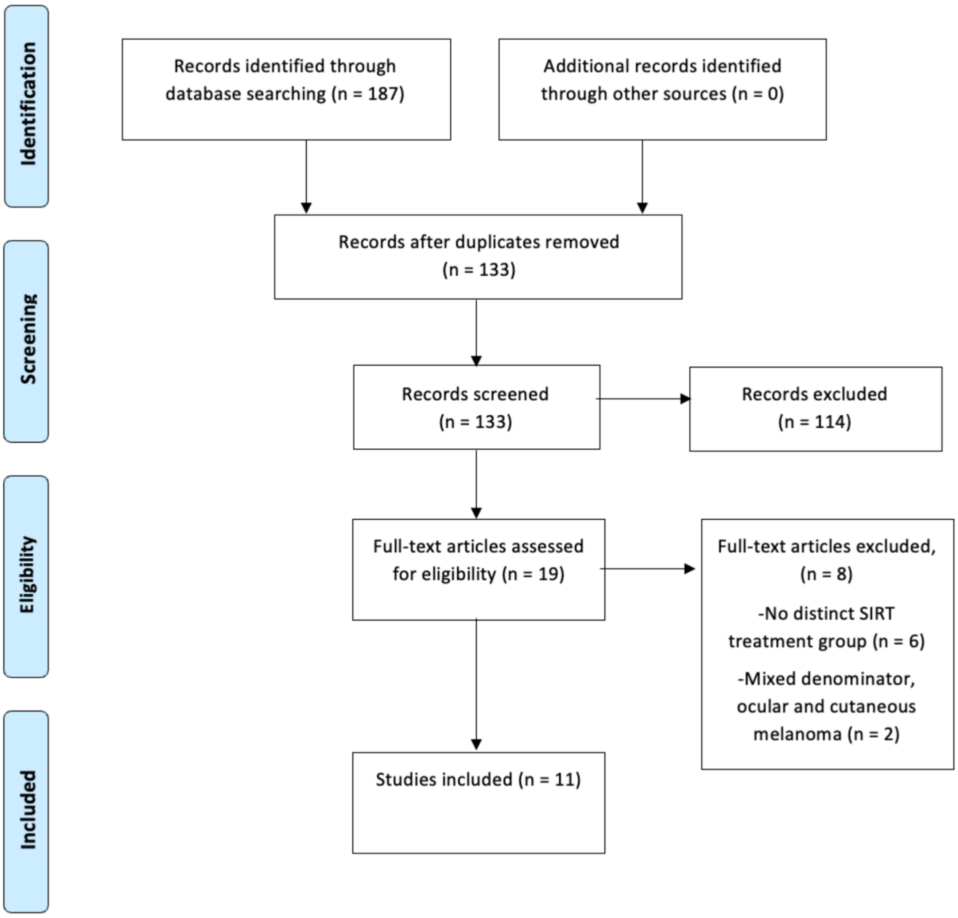
Assessment of study quality

Full text studies included in the review were assessed using the Newcastle Ottawa Scale (NOS).[24] Post-hoc quality assessment was completed by two independent reviewers (HA and DW) The NOS contains three subscales, each of which is scored separately and has different maximum scores (selection: four stars, comparability: two stars, outcome: three stars). Higher total scores represent higher quality studies with lower perceived bias.

Statistical analysis

Preliminary narrative synthesis of eligible studies was undertaken using textual descriptions and tabulation. Figures and tables were used where appropriate to facilitate ease of interpretation. Basic descriptive statistics were used to summarize data pooled from individual clinical studies. No comparative statistical analyses or tests of significance were planned or undertaken, due to study heterogeneity and a lack of randomized control trials. Estimated median OS was calculated from the median OSs reported by studies, rather than from individual patient data.

Figure 1. PRISMA diagram.[21] PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-analyses.



RESULTS

Search

A total of 187 studies were identified (Figure 1). Two non-English language articles were excluded (one case study and one review article). The full texts of 19 studies were reviewed, and 11 were deemed eligible for inclusion. No additional studies were identified from reference searches.

Study design

Study characteristics are shown in Table 1. No randomized controlled trials were identified. Most studies were single-centre (n = 9, 81.8%) and retrospective (n = 9, 81.8%). There were two prospective studies. Gonsalves et al[20] performed a prospective, two-parallel arm, Phase II trial, comparing outcomes of SIRT in treatment-naïve patients and in patients who progressed after IE. Arulananda et al[25] also performed a prospective study, examining the role of SIRT in combination with cisplatin in 10 patients.

Demographic characteristics

Demographic characteristics of study patients are shown in Table 2. The total number of patients included in the 11 studies was 268 (range 8 –71). Seventy-eight patients (29.1%) had clinical or radiological evidence of extra hepatic metastases at the time of SIRT.

The age of patients ranged from 59 to 68. Performance status was reported by 10 studies (ECOG 0–1 in four studies, ECOG 0–2 in two studies). Data on hepatic tumor load were available in seven studies (63.6%) and typically less than 50%.

Treatment

Details of SIRT therapy are shown in Table 3. 55 patients (20.5%) received systemic therapy pre-SIRT. 88 patients (32.8%) received systemic or liver-directed therapy post-SIRT.

A mean of 1.61 SIRT procedures were performed per patient (273 procedures in 170 patients). Data on the site of therapy were provided in nine studies (81.8%). Eight studies (72.7%) used resin microspheres, and a single study (9.1%) used glass microspheres or a combination of glass and resin microspheres (n = 1, 9.1%). The applied dose of radiation varied from 1.1 to 1.9 gigabecquerel (GBq).

Outcome

Data on patient outcome are provided in Table 4. Median duration of follow up was reported in six studies (54.5%), and ranged from 10 to 30 months.

10 studies (90.9%) reported liver response using the Response Evaluation Criteria in Solid Tumors (RECIST),[22] and one study

Table 1. Study design

	Year	Single/Multi center	Study design	Aim
Arulananda et al[25]	2019	Single	Prospective	To assess the feasibility of combining cisplatin with yttrium-90 microspheres in UM patients with liver-only metastases.
Eldredge-Hindy et al[18]	2016	Single	Retrospective	To report outcomes following yttrium-90 microsphere brachytherapy for unresectable liver metastases from UM and to evaluate factors predictive for overall survival and hepatic progression free survival.
Gonsalves et al[20]	2019	Single	Prospective, two-parallel arm, Phase II trial. Group A = treatment-naïve patients. Group B = participants who progressed after IE.	To evaluate, in a prospective manner, the safety and effectiveness of radioembolization for the treatment of UM hepatic metastases.
Gonsalves et al[17]	2011	Single	Retrospective	To assess the safety and efficacy of radioembolization in the management of hepatic metastasis of UM after failure of IE or CE.
Kennedy et al[26]	2009	Multi center	Retrospective	N/S
Klingenstein et al[16]	2013	Single	Retrospective	To retrospectively evaluate the overall survival, safety, and efficacy of metastatic UM patients after radioembolization as salvage therapy.
Levey et al[27]	2019	Single	Retrospective	To investigate predictors of overall survival and progression-free survival in patients with ocular melanoma metastatic to the liver undergoing yttrium-90 radioembolization, including the effect of concurrent immunotherapy.
Ponti et al[14]	2019	Single	Retrospective	To investigate the safety and efficacy of SIRT used as first-line therapy in patients with UM metastatic to the liver.
Schelhorn et al[28]	2015	Single	Retrospective	To report complications and response rates of radioembolization as salvage therapy for hepatic UM metastases.
Tulokas et al[15]	2018	Multi center	Retrospective	To retrospectively evaluate the outcome and safety of SIRT in UM patients with hepatic metastases not suitable for surgery.
Zheng et al[29]	2018	Single	Retrospective	To evaluate the safety and efficacy of yttrium-90 trans-arterial radioembolization around immunotherapy in patients with unresectable hepatic metastases from UM.

CE, Chemoembolization; IE, Immunoembolization; N/S, Not specified; SIRT, Selective internal radiation therapy; UM, Uveal melanoma.

Table 2. Patient demographics

	Number of patients	Gender (Male/ Female)	Mean age (years).	Median time to SIRT (months)	Performance status (ECOG)	Hepatic tumor load	Number of hepatic lesions	Number of patients with extra hepatic metastases
Arulananda et al[25]	10	5/5	68	3	0–1	N/S	N/S	Excluded
Eldredge-Hindy et al[18]	71	34/37	63[a]	10	KPS 70–100	Typically,<50%	N/S	39
Gonsalves et al[20]	48	22/26	59	N/S	0–1	<20% in 40 20–50% in 8	N/S	4
Gonsalves et al[17]	32	14/18	61[a]	N/S	0–2	<25% in 25 25–50% in 5 >50% in 2	N/S	Excluded
Kennedy et al[26]	11	5/6	56[a]	N/S[b]	KPS >60	N/S	>4	"Not an exclusion criterion"
Klingenstein et al[16]	13	4/9	57[a]	5	KPS >70	<25% in 3 25–50% in 8 >50% in 2	N/S	6
Levey et al[27]	24	14/10	54[a]	N/S	0–1	N/S	N/S	19
Ponti et al[14]	22	11/11	59	3	0–1	N/S	0–10 in 1 >11 in 9	4
Schelhorn et al[28]	8	5/3	59	17.1	N/S	<25% in 3 25–50% in 4 >50% in 1	N/S	1
Tulokas et al[15]	18	N/S	62	3.6	WHO 0–2	<25% in 13 25–50% in 4 N/S in 1	N/S	Excluded
Zheng et al[29]	11	4/7	66	9	0–2	<25% in 8 25–50% in 3	0–10 in 3 >10 in 8	4

ECOG, Eastern Cooperative Oncology Group; N/S, Not specified; SIRT, Selective internal radiation therapy.
[a]Median age (mean not reported).
[b]Provided time to SIRT from initial diagnosis of UM (median 25.5 months), rather than from diagnosis of hepatic metastases.

Table 3. Treatment details

	Number of patients	Pre-SIRT systemic therapy	Pre-SIRT liver-directed therapy	Number of SIRT procedures	Site of therapy	Type of microsphere	Median activity infused per patient (GBq).	Post-SIRT therapy
Arulananda et al[25]	10	Y • Anti PD-1/PD-L1 (1) • Anti CTLA-4 (1)	Y • Surgery (1)	N/S	N/S	Resin	1.8	Y • Anti PD-1 (7) • Anti CTLA-4 (1) • None (2)
Eldredge-Hindy et al[18]	71	Y • RTK inhibitor (6) • Anti CTLA-4 (2) • Other (2)	Y • IE (50) • CE (31) • Surgery (2)	134	• Unilobar (115) • Whole liver, unfractionated (5) • Whole liver, fractionated (14)	Resin	• lobe – 0.88 • lobe – 0.33 Whole liver, unfractionated – 0.84 Whole liver, fractionated – 0.59	N/S
Gonsalves et al[20]	48	N (excluded)	Y • IE (24)	N/S	• Unilobar (12) • Lobar, sequential (35) • Whole liver, fractionated (1)	Resin	Group A – 1.21 Group B – 1.30	N
Gonsalves et al[17]	32	N/S	Y • IE[a] • TACE[a] • Surgery (3)	N/S	• Unilobar (3) • Lobar, sequential (19) • Whole liver, unfractionated (8) • Whole liver, fractionated (2)	Resin	1.08	N/S
Kennedy et al[26]	11	N/S	Y • CE (1)	12	• Unilobar (3) • Whole liver (8)	Resin	1.55 per treatment	N/S
Klingenstein et al[16]	13	Y • Chemotherapy without treatment response (10)	Y • RFA (1) • Surgery (1)	16	• Unilobar (1) • Whole liver (11)	Resin	1.82 per patient	N/S
Levey et al[27]	24	Y • Concurrent immunotherapy within 3 months before or after SIRT (12)	Y • TACE (1) • MRI ablation (1)	42	• Unilobar (4) • Whole liver, fractionated (20)	Resin	N/S[b]	Y • Concurrent immunotherapy within 3 months before or after SIRT (12)

(Continued)

Table 3. (Continued)

	Number of patients	Pre-SIRT systemic therapy	Pre-SIRT liver-directed therapy	Number of SIRT procedures	Site of therapy	Type of microsphere	Median activity infused per patient (GBq).	Post-SIRT therapy
Ponti et al[14]	22	N (excluded)	N (excluded)	29	• Unilobar (8) • Whole liver, single fraction (9) • Whole liver, fractionated (5)	Glass and resin	1.8	Y • Chemotherapy (11) • Immunotherapy (13) • Both chemotherapy and immunotherapy (7) • TACE (4) • Thermal ablation (5)
Schelhorn et al[28]	8	Y • Sorafenib (6). • Dacarbazine IV (1) • BI 1247.1 (1)	Y • Intra arterial melphalan (8). • Intra arterial fotemustine (4) • TACE (1)	N/S	• Unilobar (1) • Whole liver, single fraction (6) • Whole liver, fractionated (1)	Glass	N/S[c]	Y • Treosulfan and gemcitabine (1) • Ipilimumab (1) • Dacarbazine (1) • Intra arterial melphalan (1)
Tulokas et al[15]	18	Y • Chemotherapy (4)	N/S	21	N/S	N/S	1.9	Y • Chemotherapy, surgery, immunotherapy or combination (14)
Zheng et al[29]	11	Y • Disease progression after CTLA-4 and/or PD-1 antibody therapy (9)	N/S	19	• Unilobar (4) • Whole liver, single fraction (2) • Whole liver, fractionated (5)	Resin	1.92	Y • Pembrolizumab (5) • Ipilimumab (3) • Nivolumab (2)

CE, Chemoembolization; CTLA-4, Cytotoxic T-lymphocyte-associated protein 4; GBQ, Gigabecquerel; IE, Immunoembolization; N/S, Not specified; PD-1, Programmed cell death protein 1; PD-L1, Programmed death ligand 1; RFA, Radiofrequency ablation; RTK, Receptor tyrosine kinase; SIRT, Selective internal radiation therapy; TACE, Trans-arterial chemoembolization.

[a]Provided median number of procedures and range, but not number of patients. Immunoembolization (median, 4; range 1–14 procedures) or TACE (median, 3; range, 1–14 procedures).

[b]Median not reported. Mean value reported (1.95 GBy).

[c]Median not reported. Mean value reported (4.28 GBy to right lobe and 1.51 GBy to left lobe).

Table 4. Outcome

	Number of patients	Median duration of follow-up (months)	Follow-up imaging	Liver response (RECIST criteria) (CR, PR, SD, DP)	DCR (% and proportion)	Median OS from diagnosis of liver metastases (months)	Median OS from time of SIRT (months)	Median hPFS (months)
Arulananda et al[25]	10	30	CT, MRI	0, 3, 3, 3	66.7% (6/9)	N/S	10	3
Eldredge-Hindy et al[18]	71	9.9	CT, MRI, PET	0, 5, 32, 24[a]	60.7% (37/61)	23.9	12.3	5.9
Gonsalves et al[20]	48	"For at least 2 years or until death"	CT, MRI, PET	0, 17, 17, 13 Group A – 0, 9, 11, 3 Group B – 0, 8, 6, 10	72.3% (34/47)	N/S	N/S Group A – 18.5 Group B – 19.2	N/S Group A – 8.1 Group B – 5.2
Gonsalves et al[17]	32	10.0	CT, MRI, PET	1, 1, 18, 12	62.5% (20/32)	N/S	10	4.7
Kennedy et al[26]	11	"Follow-up continuing on all survivors" at time of publication	CT, MRI, PET	1, 6, 1, 1	88.9% (8/9)	N/S	N/S	N/S
Klingenstein et al[16]	13	N/S	CT, MRI, PET	0, 8, 2, 3	76.9% (10/13)	19	7	N/S
Levey et al[27]	24	N/Sb	CT, MRI	0, 2, 11, 9 Group A – 0, 1, 6, 5 Group B – 0, 1, 5, 4	59.1% (13/22)	26.3	18.6 Group A – 26 Group B – 9.5	5.2 Group A – 10.3 Group B – 2.7
Ponti et al[14]	22	15	CT/MRI	0, 0, 21, 4	84.0% (21/25)	20	18	8
Schelhorn et al[28]	8	N/S	CT	0, 0, 3, 4	42.9% (3/7)	19.9	2.8 (12.3 weeks)	1.0 (4.3 weeks)
Tulokas et al[15]	18	13.5	CT, MRI, PET	0, 3, 8, 5	68.8% (11/16)	19.6	13.5	5.6
Zheng et al[29]	11	10.5	CT, MRI	1, 2, 4, 4[c]	63.6% (7/11)	33.5	17	15

CR, Complete response; DCR, Disease control rate; DP, Disease progression; N/S, Not specified; OS, Overall survival; PR, Partial response; RECIST, Response evaluation criteria in solid tumors; SD, Stable disease; SIRT, Selective internal radiation therapy; hPFS, Hepatic progression free survival.

[a]Data also reported using PET in 29 patients (CR 0, PR 10, SD 14, DP 5).

[b]Median value not reported. Mean value reported (43 months).

[c]RECIST criteria not used. mRECIST criteria reported.

(9.1%) used modified RECIST criteria.[30] Disease control rate (DCR) was calculated as the sum of complete response (CR), partial response (PR) and stable disease (SD). Liver response data were available in 252 patients, and disease control was achieved in 170 (67.5%).

Median OS (from time of SIRT) was reported in nine studies, and ranged from 2.8 to 18.6 months (median 12.3 months). Median OS (from diagnosis of hepatic metastases) was also reported in seven studies, and ranged from 19 to 33.5 months (median 20 months). Median hPFS was reported in eight studies, and ranged from 1 to 15 months (median 5.4 months).

Toxicity

Toxicities were graded according to the CTCAE (v. 3 or v. 4) in 10 studies (90.9%). A single study (9.1%) used the CIRSE classification system.[27,31] Low-grade (CTCAE Grade 1–2) clinical complications after SIRT were common, including abdominal pain, fatigue and nausea/vomiting (Table 5). High grade complications (CTCAE Grade 3 or above) were rare, but included radiation cholecystitis ($n = 3$), liver failure ($n = 3$), gastric ulceration ($n = 2$), pain ($n = 2$), nausea/vomiting ($n = 1$) and hepatomegaly ($n = 1$). One hepatic artery dissection was reported, but the severity was not graded.[15]

There were two SIRT-related deaths; both were secondary to liver failure. The first occurred in a patient with arterioportal venous shunting on diagnostic angiography, but had no therapeutic alternatives to SIRT.[16] The second patient experienced liver failure and Budd-Chiari syndrome post-SIRT. It was unclear whether the patient decompensated from radioembolization-induced liver injury or liver injury secondary to disease progression, however a diagnosis of radiation-induced liver disease was made.[27]

Derangement of liver function tests (LFTs) was common post-SIRT, and typically CTCAE Grade 1–2 (Table 5). Grade 3–4 biochemical toxicity was less common ($n = 17$).

Assessment of study quality

Assessment of methodological quality of the included studies is summarized in Supplementary Material 2. The NOS for cohort studies was used for assessment.[24] Overall study quality was deemed to be moderate (median total score = 6, interquartile range = 1), with some risk of bias. This review examined the safety and efficacy of SIRT for patients with liver metastases from UM. One study included a historical comparison group, however the investigators did not adjust for confounding variables.[15] The remaining studies also scored low in the comparability subscale. Scores in the outcome subscale were variable. Three studies (27.3%) did not report follow-up duration. Two studies (18.2%) had more than 10% of patients lost to follow-up and were therefore deemed to be at some risk of bias.

DISCUSSION

This systematic review evaluated the safety and efficacy of SIRT in 268 patients with liver metastases from UM. We have demonstrated that SIRT confers a DCR of 67.5% and a median OS of 12.3 months from SIRT. Median OS (from diagnosis of hepatic metastases) was 20 months and median hPFS was 5.4 months. Mild side-effects were common, including pain, fatigue, nausea/vomiting and LFT derangement. More serious side-effects, such as gastric ulceration and liver failure were rare. There were two treatment-related deaths.

Metastatic UM is a rare but aggressive malignancy. The development of hepatic metastases is an important clinical determinant of survival in patients with UM.[4] Unfortunately, there are no evidence-based guidelines for the management of hepatic metastases from UM. In patients with resectable disease, hepatic resection is the treatment of choice. Unfortunately, the majority of patients present with disease that is beyond resectability. Systemic therapies are typically ineffective in UM. As a result, a variety of liver-directed therapies have been described for UM metastases, including surgery and hepatic arterial-based treatments (TACE, IHP, HAI and IE).[8]

Previous studies have shown variable survival outcomes in UM patients treated with liver-directed therapies. A previous systematic review by Rowcroft et al[8] identified 10 studies reporting outcomes in patients undergoing surgery for UM hepatic metastases, with OS rates ranging between 10 and 35 months. The same review identified 10 studies utilizing IHP (OS: 9–25 months), eight studies evaluating HAI (OS: 10–24 months) and 17 studies reporting outcomes after TACE (OS: 5–29 months). The survival results for SIRT identified in our systematic review are comparable to those described by Rowcroft et al[8]. These results suggest that SIRT may be an appropriate treatment for some patients with hepatic metastases from UM. The additional benefits of SIRT include that it is minimally invasive, usually well-tolerated, and may be an option for patients who are unfit for open surgery.

Most studies included in our review were small and retrospective. There was significant heterogeneity in patient demographics (including tumor-load), treatment combinations (both pre- and post-SIRT), follow-up (duration and imaging modalities) and outcome measures. The median NOS score was 6, indicating a moderate risk of bias. Heterogeneous study designs limited our ability to conduct a meta-analysis, or to determine the impact of various risk factors on survival (including site of therapy, timing of therapy or combination of SIRT with other liver-directed or systemic therapies). Another limitation of the current study was the exclusion of studies reporting outcomes in cohorts with a mixed denominator (cutaneous and uveal melanoma).[32–34] We also excluded conference abstracts and unpublished data which may have led to publication bias. Whilst a risk of bias assessment was undertaken, we did not formally investigate inconsistency, indirectness, imprecision or publication bias, so the certainty of our evidence is unclear.

The optimal timing of SIRT in patients with hepatic metastases from UM is unclear. Many studies included in this review reported the use of SIRT as salvage therapy (following failure of various systemic or liver-directed therapies). One prospective study analyzed SIRT when used as first-line *vs* as salvage therapy.[20] The authors showed a DCR of 87% in 24 treatment-naïve patients

Table 5. Toxicity

	Number of patients.	Clinical complications (all grade CTCAE 1 or 2 unless otherwise specified)				Biochemical complications (all CTCAE Grade 1 or two unless otherwise specified)		SIRT-related deaths
		Fatigue	Nausea or vomiting	Abdominal pain	Other clinical complications	LFT derangement	Other	
Arulananda et al[25]	10	4	1	5	Anorexia: 2	N/S	N/S	0
Eldredge-Hindy et al[18]	71	31	11	18	N/S	34 (including 5 patients with Grade 2 or 3). (17 had pre-existing Grade 1–2 LFT dysfunction prior to treatment)	N/S	0
Gonsalves et al[20]	48	39	28 (including one patient with Grade 3)	20	Fever: 1 Weight loss: 2 Pain: 16 (including one patient with Grade 3) Anorexia: 7 Bloating/flatulence: 8 Diarrhea/ constipation: 10 Dysgeusia: 6 Dyspepsia: 6 Dysphagia: 3 GERD: 10 Gastritis: 1 Back pain: 9 Leg/shoulder pain: 6 (including one patient with Grade 3) Bruising or hematoma: 6 Shortness of breath: 2	45 (including one patient with Grade 3)	Increase in total bilirubin: 3 Decrease in lymphocytes: 11 (including 2 patients with Grade 3)	0
Gonsalves et al[17]	32	9	N/S	5	Indigestion: 2	18 (including 3 patients with Grade 3 and 1 patient with Grade 4 liver toxicity, deemed to be caused by tumor progression and not related to the treatment itself) (13 had pre-existing Grade 1–3 LFT dysfunction prior to treatment)	N/S	0

(Continued)

Table 5. (Continued)

	Number of patients.	Clinical complications (all grade CTCAE 1 or 2 unless otherwise specified)				Biochemical complications (all CTCAE Grade 1 or two unless otherwise specified)		SIRT-related deaths
		Fatigue	Nausea or vomiting	Abdominal pain	Other clinical complications	LFT derangement	Other	
Kennedy et al[26]	11	N/S	N/S	N/S	Gastric ulcer: 1 (Grade 3).	"Asymptomatic rise in bilirubin in most patients but remaining below the upper limit of normal"	N/S	0
Klingenstein et al[16]	13	N/S	3	5	Gastric ulcer: 1 Hepatomegaly: 1 (Grade 4)	N/S	N/S	1 – liver failure.
Levey et al[27]	24	N/S	N/S	N/S	Liver failure: 1 (CIRSE Grade 6)	N/S	N/S	1 – liver failure
Ponti et al[14]	22	N/S*	N/S	N/S[a] (2 patients with Grade 3)	Radiation cholecystitis: 3 (Grade 3) Liver failure: 2 (Grade 3)	N/S	Laboratory perturbations: 16 (including 4 patients with Grade 3 and 3 patients with Grade 4)	0
Schelhorn et al[28]	8	N/S	1	N/S	N/S	N/S	0 "No new relevant radioembolization-induced laboratory toxicity"	0
Tulokas et al[15]	18	N/S[b]	N/S[b]	N/S[b]	Hepatic artery dissection: 1[c] Fever in relation to liver toxicity: 1	13 (includes one patient with Grade 3 and 1 patient with Grade 4 elevation of transaminases)	N/S	0
Zheng et al[29]	11	3	0	3	Peptic ulcer with hemorrhage (Grade 3), hemi gastrectomy performed after 6 months.	7 (incudes one patient with Grade 3 elevation of ALT, AST, and total bilirubin, felt to be related to progression of liver metastases)	N/S	0

ALT, Alanine aminotransferase; AST, Aspartate aminotransferase; CIRSE, Cardiovascular and Interventional Radiological Society of Europe; CTCAE, Common terminology criteria for adverse events; GERD, Gastro-esophageal reflux disease; LFT, Liver function test; N/S, Not specified; SIRT, Selective internal radiation therapy.

[a]Individual numbers not provided. Nine patients experienced Grade 1 or 2 abdominal pain or fatigue.

[b]Individual numbers not provided. 15 patients experienced nausea, abdominal pain, fatigue or sub febrility.

[c]Grade not provided.

and a DCR of 58.3% in patients who progressed after IE. This suggests that SIRT may be effective both as first-line and salvage therapy in patients with metastatic UM.

Similarly, the optimal combination of SIRT with other systemic or liver-directed therapies is unclear. No randomized controlled trials have examined the role of SIRT in combination with other therapies in the treatment of hepatic metastases from UM. Levey et al[27] examined the effect of concomitant immunotherapy on OS in a retrospective cohort of patients treated with SIRT. Patients treated with concomitant immunotherapy had prolonged survival compared to patients treated with SIRT alone (26 *vs* 9.5 months). This difference was statistically significant, suggesting synergistic and enhanced anti tumor effect when immunotherapy and SIRT are combined. Similarly, Zheng et al[29] reported promising results with the combination of SIRT and immunotherapy, although their study was small and retrospective. There is an increasing body of evidence suggesting that radiation therapy can augment the anti tumor effect of immune checkpoint blockade therapies by facilitating tumor-antigen presentation to immune effector cells.[35]

Future work should focus on prospective studies and randomized controlled trials to further define the role of SIRT in hepatic metastases from UM. Two trials are currently examining the role of SIRT in combination with sorafenib and ipilimumab/nivolumab (NCT01893099 and NCT02913417, respectively). Another Phase II randomized controlled trial is currently comparing SIRT versus TACE with cisplatin in UM (NCT02936388).

SIRT is expensive, and an evaluation of heath economic cost should also be sought before this treatment gains regulatory approval for use in metastatic UM. One previous study has shown that SIRT is a cost-effective therapy for patients with metastatic colorectal cancer, when compared with best-supportive care.[36] However, no study included in this review outlined the health economic cost of SIRT for patients with UM. There are strong arguments for robust trials comparing SIRT with other treatments for metastatic UM, and these studies should also include prospective cost–benefit analysis.

Consistent measurement and reporting of outcomes is also necessary to robustly evaluate new technology, and there is a strong case for developing a set of reporting standards for use in SIRT trials. On the basis of the current review, we recommend a combination of patient demographic characteristics, treatment details and outcome measures be included in future SIRT trials for UM (Supplementary Material 3). Patient characteristics should include demographic characteristics, hepatic tumour load and the presence of extra hepatic disease. Treatment details should include concurrent treatment, site of therapy and number of treatments. Outcome measures of interest include liver response (using RECIST criteria), OS, hPFS, and clinical/biochemical complications (using CTCAE). A measure of health-economic cost should also be considered for inclusion. Consultation of treatment providers and patients is necessary for development of a core outcome set for future SIRT trials.

In conclusion, SIRT appears to be a safe and effective treatment for patients with unresectable liver metastases from UM, but our results were limited by heterogeneity in study design and small patient numbers. Further prospective studies and randomized controlled trials are required to explore the role of SIRT in patients with UM hepatic metastases.

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